

MACKALL AAF, NC, USA

WMO: 723034

Lat: 35.033N

Lon: 79.500W

Elev: 115

StdP: 99.96

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 93747

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.8	(c) -5.6	(d) -17.4	(e) 0.8	(f) -2.3	(g) -14.6	(h) 1.1	(i) -1.7	(j) 10.2	(k) 17.3	(l) 8.7	(m) 12.3	(n) 1.5	(o) 340	(p) 0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	35.2	24.1	34.1	23.8	32.9	23.6	26.4	31.3	25.6	30.8	25.0	30.2	3.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	20.8	27.9	24.1	19.3	27.3	23.7	18.7	26.9	82.5	31.3	78.9	30.8	76.3	30.4	31.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.3	(b) 7.3	(c) 6.3	(e) -12.2	(f) 37.1	(g) 3.6	(h) 1.6	(i) -14.8	(j) 38.2	(k) -16.9	(l) 39.1	(m) -18.9	(n) 40.0	(o) -21.5	(p) 41.1		
(a) 8.3	(b) 7.3	(c) 6.3	(e) -12.7	(f) 26.6	(g) 3.6	(h) 0.5	(i) -15.3	(j) 27.0	(k) -17.4	(l) 27.3	(m) -19.4	(n) 27.6	(o) -22.0	(p) 28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	4.8	7.7	10.9	16.8	21.4	25.6	27.1	26.1	23.2	16.9	9.8	7.8		
	DBStd	8.87	5.87	5.62	5.39	4.37	3.46	2.53	1.91	2.11	2.98	4.53	4.56	5.45		
	HDD10.0	517	182	103	57	3	0	0	0	0	0	3	59	110		
	HDD18.3	1719	421	299	238	79	14	0	0	0	3	81	256	327		
	CDD10.0	2916	19	39	83	208	355	469	531	498	397	218	54	43		
	CDD18.3	1076	1	2	7	34	111	219	273	240	149	37	1	2		
	CDH23.3	10205	2	18	71	344	992	2233	2723	2182	1285	341	8	6		
	CDH26.7	4318	0	1	7	83	346	1046	1281	945	520	90	0	0		
(14) Wind	WSAvg	2.6	2.5	3.0	3.1	3.3	2.9	2.6	2.5	2.1	2.3	2.4	2.3	2.3		
(15) Precipitation	PrecAvg	1188	94	84	105	81	78	116	130	127	128	84	79	81		
	PrecMax	1582	170	163	186	155	184	317	224	253	376	232	213	172		
	PrecMin	837	22	41	31	26	32	20	48	32	33	0	19	39		
	PrecStd	164	45	36	41	34	35	72	55	49	82	60	52	33		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.9	25.7	27.8	30.8	34.3	37.3	36.7	35.3	34.5	31.6	24.7	24.2		
		MCWB	16.4	17.8	16.8	19.2	22.3	24.1	24.5	24.1	21.9	23.1	17.7	19.7		
	2%	DB	19.4	22.9	25.6	29.1	31.8	35.1	35.2	34.1	32.8	28.9	22.6	21.6		
		MCWB	15.4	16.1	16.6	18.7	21.0	23.7	24.7	24.1	22.5	21.5	17.2	18.5		
	5%	DB	17.2	20.2	23.3	27.2	30.3	33.7	34.0	32.8	31.5	26.9	20.7	19.6		
		MCWB	13.5	14.7	15.1	18.2	20.7	23.2	24.6	24.0	22.7	20.6	16.1	17.4		
	10%	DB	14.7	17.7	20.6	25.3	28.6	32.1	32.7	31.5	30.0	24.9	18.7	17.3		
		MCWB	11.5	13.4	14.5	17.1	20.4	22.9	24.3	23.7	22.5	19.9	14.7	14.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.7	19.2	19.1	22.2	23.9	25.8	28.6	26.3	29.9	25.8	20.6	21.0		
		MCDB	21.0	23.2	23.7	26.4	29.0	32.7	31.8	32.1	30.6	27.0	22.0	22.2		
	2%	WB	16.7	17.8	18.3	21.0	22.8	25.1	26.7	25.7	25.6	23.9	19.0	19.3		
		MCDB	18.8	21.0	22.8	25.2	27.6	31.8	31.7	31.0	28.2	26.2	20.9	20.9		
	5%	WB	14.4	16.3	17.2	19.9	22.1	24.4	25.7	25.2	24.2	22.7	17.4	17.7		
		MCDB	16.1	19.4	20.9	24.1	27.4	30.8	31.2	30.1	28.3	26.0	19.4	19.1		
	10%	WB	11.6	13.9	16.0	18.8	21.5	23.8	25.2	24.6	23.4	21.1	15.6	15.4		
		MCDB	14.0	17.3	19.5	23.5	26.5	29.7	30.6	29.3	27.3	23.6	18.1	17.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	12.5	12.8	13.5	12.1	11.8	10.9	10.4	11.6	12.9	13.6	11.8		
		MCDBR	14.8	15.1	15.7	15.1	14.1	13.4	12.3	11.9	13.8	14.3	15.4	12.5		
		MCWBR	10.7	9.4	7.7	6.5	5.0	4.1	3.9	3.7	5.3	7.4	10.1	9.3		
		MCDBR	12.4	13.4	12.6	11.3	11.1	12.0	11.0	10.4	10.2	11.5	11.3	10.9		
	5% WB	MCWBR	10.7	9.3	7.4	5.8	4.5	4.2	4.5	3.6	5.6	7.9	9.4	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.327	0.362	0.401	0.459	0.490	0.538	0.513	0.427	0.389	0.335	0.312		
	taud		2.502	2.462	2.396	2.283	2.165	2.120	2.005	2.064	2.300	2.373	2.484	2.545		
	Ebn at Noon		892	919	909	885	834	805	763	775	831	835	859	874		
	Edh at Noon		83	96	112	132	150	157	175	162	122	105	84	76		
(44) All-Sky Solar Radiation	RadAvg		2.66	3.31	4.39	5.50	6.04	6.38	6.04	5.48	4.65	3.81	2.91	2.31		
	RadStd		0.16	0.38	0.33	0.39	0.49	0.45	0.38	0.26	0.40	0.49	0.23	0.19		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (71 neighbors)	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+117

Nomenclature: See separate page